

Xiaoyang Cao

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EDUCATION

Massachusetts Institute of Technology

M.S. in Computational Science and Engineering (CSE)

M.S. in Technology and Policy (TPP)

Cambridge, MA

Sep 2025 – May 2027 (expected)

Sep 2025 – May 2027 (expected)

Tsinghua University

B.S. in Mathematics and Physics

Beijing, China

Sep 2021 – Jun 2025

- GPA: 3.95 / 4.00 — ranked 3 / 60 (top 5%)
- Coursework: Operations Research (A+), Quantum Mechanics (A+), Fundamentals of Physics (A+), Numerical Analysis (A), Principles of Economics (A)

University of California, Berkeley

Exchange Student — GPA: 3.93 / 4.00

Berkeley, CA

Jan 2024 – Aug 2024

- Coursework: EECS 127 Optimization Models in Engineering (A+), CS 188 Introduction to Artificial Intelligence (A+)

PUBLICATIONS

Xiaoyang Cao, Zelai Xu, Mo Guang, Kaiwen Long, Michiel Bakker, Yu Wang, Chao Yu. “RE-PO: Robust Enhanced Policy Optimization as a General Framework for LLM Alignment.” *International Conference on Learning Representations (ICLR)*, 2026.

Xiaoyang Cao, Zhe Fu, Alexandre M. Bayen. “Pareto Control Barrier Function for Inner Safe Set Maximization Under Input Constraints.” *American Control Conference (ACC)*, 2025.

Xiaoyang Cao, Dingyi Zhuang, Shenhao Wang, Jinhua Zhao. “Virtual Nodes Improve Long-term Traffic Prediction.” *Transportation Research Board Annual Meeting (TRB)*, 2025.

RESEARCH EXPERIENCE

Robust Preference Alignment for LLMs (ICLR 2026)

Mar 2025 – Sep 2025

Advised by Prof. Yu Wang, Tsinghua University

- Proposed RE-PO, an EM-based preference-alignment framework that treats the correctness of each preference label as a latent variable, jointly infers label and annotator reliability, and adaptively re-weights preference pairs to mitigate noisy feedback.
- Generalized the framework to arbitrary preference losses via a Gibbs-distribution connection to probabilistic models, enabling systematic robustification of DPO, IPO, SimPO, and CPO; proved convergence to the true noise level under a calibrated model.
- Improved AlpacaEval 2 win rates by up to +7.0 pp on Mistral-7B and Llama-3-8B backbones, with further gains on Arena-Hard and the real multi-annotator dataset MultiPref. [arXiv:2509.24159](https://arxiv.org/abs/2509.24159)

Safe Control via Pareto Control Barrier Functions (ACC 2025)

May 2024 – Oct 2024

Advised by Prof. Alexandre Bayen, UC Berkeley

- Developed the Pareto Control Barrier Function (PCBF) algorithm by integrating Pareto multi-task learning into neural control barrier functions, maximizing the inner safe set while guaranteeing system safety under input constraints.
- Matched the Hamilton–Jacobi viability kernel on a 2D inverted pendulum and achieved a 700% larger inner safe set than the benchmark on a 12-D quadrotor obstacle-avoidance task. [IEEE Xplore](#)

RESEARCH EXPERIENCE (CONTINUED)

Virtual Nodes for Long-term Traffic Prediction (TRB 2025)

May 2024 – Sep 2024

Advised by Prof. Jinhua Zhao, MIT

- Introduced virtual nodes into spatio-temporal graph neural networks to mitigate over-squashing in long-term traffic prediction, constructing a semi-adaptive adjacency matrix from learnable node embeddings.
- Reduced RMSE by 6.27% and MAPE by 5.04% at long horizons on a five-year, 716-sensor San Diego dataset; the learned adjacency assigns higher weights to traffic-intensive intersections, improving interpretability.
- Full manuscript in preparation for *Transportation Research Part C: Emerging Technologies* [arXiv:2501.10048](#)

Travel Behavior Prediction from Satellite Imagery and Socioeconomic Data

Jun 2023 – Dec 2023

Advised by Prof. Jinhua Zhao, MIT

- Extracted latent features from 30,000+ satellite images of Chicago using VQ-GAN and fused them with socioeconomic indicators via LASSO, improving prediction accuracy by 10% across three types of travel behavior.
- Resolved codebook collapse by applying k-means clustering to the VQ-GAN codebook, reducing its size from 1,024 to 8 entries while raising utilization from 9.5% to 100%.

Urban Traffic Accident Duration via Deep Survival Analysis

Sep 2022 – Jan 2024

Advised by Prof. Ruimin Li, Tsinghua University

- Developed DeepCAFT, a deep competing-risk accelerated failure time model that learns a generalized hazard function to capture non-linear relationships between accident duration and influencing factors, relaxing the unrealistic i.i.d. assumption of classical survival models.
- Achieved a 9% MAPE improvement over traditional models on 5,386 accident records from Beijing's 4th Ring Road; under review at *IET Intelligent Transport Systems*.

AWARDS & HONORS

Outstanding Winner & INFORMS Award, Mathematical Contest in Modeling — top 1% worldwide	May 2023
Outstanding Innovation in Science & Technology Scholarship, Tsinghua University — top 10–15%, awarded twice	2023, 2024
Comprehensive Excellence Scholarship, Tsinghua University — top 10% for academics, leadership, and service	Oct 2023
Outstanding Academic Scholarship, Tsinghua University — top 10% GPA	Oct 2022

SKILLS & LANGUAGES

Programming: Python (PyTorch, scikit-learn, NumPy, Pandas), R (tidyverse, ggplot2, survival analysis, time-series forecasting), MATLAB

Optimization & Simulation: Gurobi, CVXPY, SciPy, Simulink; Monte Carlo simulation and dynamic-systems modeling for nonlinear control and real-time optimization

Languages: English (fluent), Chinese (native)